
Decompose and Recompose: Reasoning New Skills from Existing Abilities for Cross-Task Robotic Manipulation

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Abstract

Cross-task generalization is a core challenge in open-world robotic manipulation, and the key lies in extracting transferable manipulation knowledge from seen tasks. Recent in-context learning approaches leverage seen task demonstrations to generate actions for unseen tasks without parameter updates. However, existing methods provide only low-level continuous action sequences as context, failing to capture composable skill knowledge and causing models to degenerate into superficial trajectory imitation. We propose *Decompose and Recompose*, a skill reasoning framework using atomic skill–action pairs as intermediate representations. Our approach decomposes seen demonstrations into interpretable skill–action alignments, enabling the model to recompose these skills for unseen tasks through compositional reasoning. Specifically, we construct a task-adaptive dynamic demonstration library via visual-semantic retrieval combined with skill sequences from a planning agent, complemented by a coverage-aware static library to fill missing skill patterns. Together, these yield skill-comprehensive demonstrations that explicitly elicit compositional reasoning for skill composition and execution ordering. Experiments on the AGNOSTOS benchmark and real-world environments validate our method’s zero-shot cross-task generalization capability.

1. Introduction

With the rapid development of Vision-Language-Action (VLA) models (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024; Team et al., 2024), robotic manipulation has achieved significant progress in handling visual perturba-

tions within known tasks. However, real-world deployment inevitably involves novel objects, new goals, and action compositions that were never observed. (Zhou et al., 2025) introduces the cross-task generalization setting, which requires robots to transfer manipulation knowledge from seen tasks to entirely unseen tasks without parameter updates, as illustrated in Figure 1(a). Compared to within-task generalization that focuses on visual robustness under fixed task semantics, cross-task generalization requires transferable skill understanding, which is essential for deploying general-purpose robots in open-world environments.

The core challenge of cross-task generalization lies in extracting transferable manipulation knowledge from seen tasks and reasoning over unseen tasks. Existing methods such as X-ICM (Zhou et al., 2025) train a diffusion-based dynamics-guided sample selection module to retrieve demonstrations from seen tasks based on dynamics similarity, and use them to prompt LLMs for action prediction. However, these methods require training on specific task distributions, weakening cross-domain transferability, and provide only low-level numerical action sequences as context, relying on direct alignment of trajectory shapes without explicit skill semantics. Such numerical representations fail to express causal and procedural information, including what skill is being performed at each step, why it is executed, and how it connects to subsequent steps. This causes LLMs to only perform trajectory pattern matching rather than skill reasoning, limiting their generalization capability to tasks with novel skill compositions.

To address these issues, as shown in Figure 1(b), we propose *Decompose and Recompose*, a compositional skill reasoning framework that reasons new skills from existing abilities for cross-task manipulation generalization. We decompose each task demonstration into atomic skill sequences and atomic skill label–action alignment pairs, where the former enables skill sequence similarity retrieval and reasoning, and the latter provides composable skill structures for inference, while maintaining executable low-level actions as the final output. This elevates cross-task transfer from trajectory shape similarity to composable skill structures, explicitly activating the model’s reasoning capabilities for skill composition and execution ordering.

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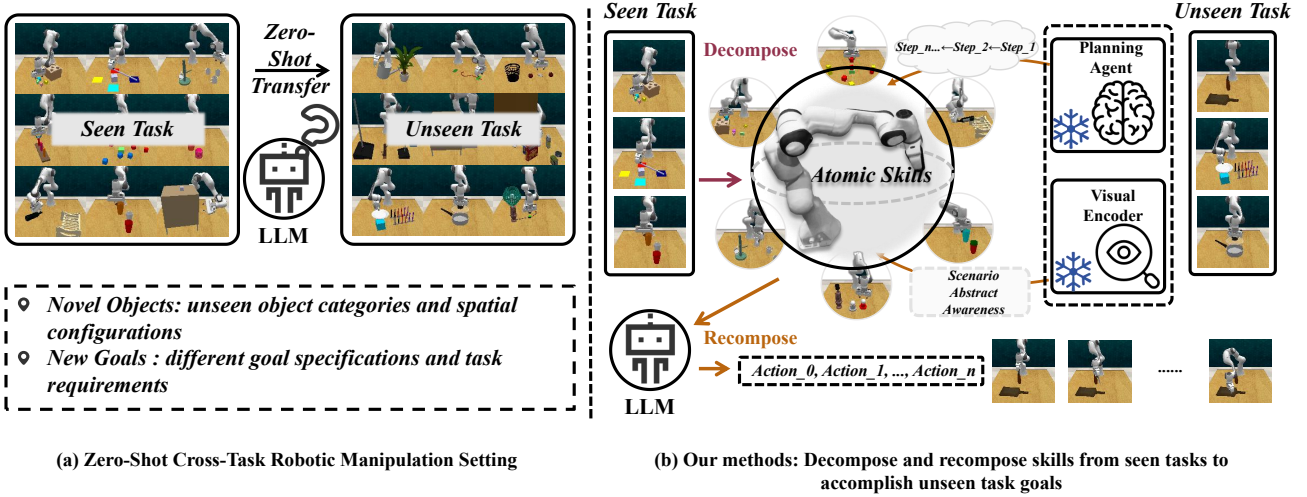


Figure 1. Overview of cross-task robotic manipulation. (a) The zero-shot cross-task setting requires transferring knowledge from seen tasks to unseen tasks involving novel objects and new goals. (b) Our *Decompose and Recompose* framework extracts atomic skills from seen task demonstrations, uses a planning agent to predict skill sequences for unseen tasks, and leverages visual encoding for scene-aware retrieval. The LLM then reasons over skill–action aligned demonstrations to generate executable action sequences for the target task.

Specifically, we employ a general vision model (Siméoni et al., 2025) to extract static visual-semantic representations for scene similarity retrieval, and use a planning agent to predict atomic skill sequences for unseen tasks. Based on these predicted sequences, we retrieve and rerank demonstrations with similar action logic from the candidate pool, constructing a task-adaptive dynamic demonstration library. However, relying solely on dynamic retrieval may still result in incomplete demonstration coverage: even if the retrieved demonstrations have high overall similarity, they may lack critical skill patterns required for task completion. To address this, we introduce an offline-constructed coverage-aware static demonstration library, converting planner-output skill sequences into structured tokens (e.g., verb skills, skill chains) and normalizing the skill token space through Inverse Document Frequency (IDF) based Token Weighting and Length Regularization strategies. During inference, we compute the covered token set from the dynamic library, explicitly identify gaps, and greedily supplement missing tokens from the static library. This yields demonstrations that are both task-adaptive and skill-coverage sufficient, making in-context demonstrations a more helpful skill evidence collection for unseen tasks, thereby enhancing cross-task compositional generalization.

In summary, our main contributions are as follows:

- We construct atomic skill label-action aligned data, providing LLMs with composable and interpretable atomic skill sets that enable structured reasoning over manipulation sequences.
- We propose *Decompose and Recompose*, a compositional skill reasoning framework for zero-shot cross-

task manipulation that obtains task-adaptive and skill-comprehensive demonstration sets through the synergy of a dynamic demonstration library for task relevance and a static demonstration library for skill coverage.

- We validate our method’s effectiveness on cross-task zero-shot generalization toward unseen tasks on the AGNOSTOS benchmark and real-world environments, achieving competitive performance.

2. Related Works

Vision-Language-Action Models. Vision-Language-Action (VLA) models enable robots to understand instructions and interact with the physical world, achieving significant progress in general-purpose manipulation. Existing VLA models primarily follow two paradigms. Modular approaches decompose perception, language understanding, planning, and action execution into separate components, such as VoxPoser (Huang et al., 2023), MOKA (Liu et al., 2024a), COPA (Huang et al., 2024a), and ReKep (Huang et al., 2024b), which typically require task-specific prompt engineering or hand-crafted designs. End-to-end approaches train policy models to directly map raw sensory inputs to robot actions, with success heavily dependent on training data scale and diversity. Representative works include RT-2 (Brohan et al., 2023) and RT-2-X (Vuong et al., 2023) trained on large-scale robotic datasets, OpenVLA (Kim et al., 2024) as the first fully open-source VLA model, and recent advances like π_0 (Black et al., 2024), RDT (Liu et al., 2024b), LLARVA (Niu et al., 2024), and HPT (Wang et al., 2024) that incorporate sophisticated architectures and training objectives. Despite progress in visual robustness within

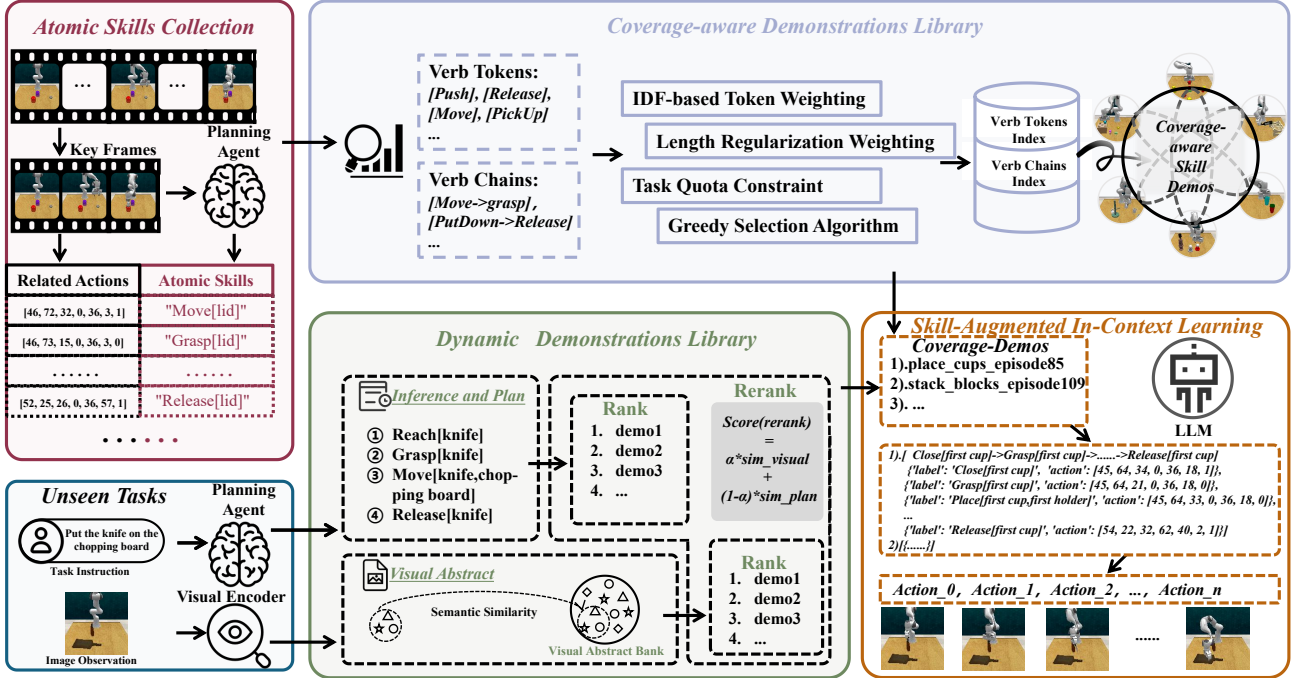


Figure 2. Overview of our method. Our framework consists of four components: (1) **Atomic Skills Collection** extracts skill–action pairs from seen demonstrations as composable intermediate representations; (2) **Coverage-aware Static Library** uses IDF-based token weighting to ensure skill pattern coverage; (3) **Dynamic Demonstrations Library** retrieves task-adaptive examples via visual and plan-based similarity; (4) **Skill-Augmented In-Context Learning** feeds retrieved demonstrations to an LLM for compositional skill reasoning to predict action sequences.

known tasks, existing VLA models remain limited in zero-shot cross-task generalization. AGNOSTOS (Zhou et al., 2025) provides the first systematic evaluation, revealing widespread degradation on unseen tasks and indicating that large-scale pre-training alone is insufficient for cross-task transfer.

In-Context Learning. Large language models demonstrate remarkable in-context learning (ICL) capabilities (Brown et al., 2020), generalizing to new tasks from few demonstrations without parameter updates. ICL has been successfully applied across various domains (Dong et al., 2022; Wies et al., 2023; Rubin et al., 2021; Min et al., 2022; Zhang et al., 2023). In robotics, RoboPrompt (Yin et al., 2024) first showed that text-only LLMs can directly predict robot actions through ICL by converting episodes into textual descriptions. Subsequent works like KAT (Di Palo & Johns, 2024), InCoRo (Zhu et al., 2024), and Instant Policy (Vosylius & Johns, 2025) further explored ICL for robotic control, but primarily focused on within-task generalization. Since demonstration quality significantly affects ICL performance (Rubin et al., 2021; Min et al., 2022), and long contexts may cause degradation (Dong et al., 2022; Wies et al., 2023), selecting effective examples is crucial. Recently, X-ICM (Zhou et al., 2025) first extended ICL to the zero-shot cross-task setting with dynamics-guided sample

selection. However, it provides only low-level action sequences as context, which lack explicit composable skill structures and causal information, causing LLMs to degrade into surface-similarity-based trajectory imitation. In contrast, our method constructs aligned representations of atomic skill labels and low-level actions, providing composable and interpretable intermediate structures that elevate cross-task transfer from “trajectory similarity” to “skill composability,” explicitly activating reasoning over skill composition and execution order.

3. Method

We propose a framework for zero-shot cross-task robotic manipulation that addresses the fundamental challenge of transferring operational knowledge from seen tasks to unseen tasks. Unlike prior approaches that rely on training dynamics-guided retrievers on specific task distributions, our method achieves cross-task generalization through a training-free pipeline that leverages pretrained visual encoders and large language models. The core insight is that providing LLMs with composable and interpretable intermediate representations—atomic skill labels aligned with low-level actions—is more effective for cross-task transfer than simply presenting raw numerical action sequences.

As illustrated in Figure 2, our approach consists of three tightly integrated components: (1) **atomic skill collection** that constructs label-action aligned data for seen task demonstrations; (2) **dual-library demonstration retrieval** that combines visual scene similarity with plan-based skill coverage; and (3) **skill-augmented in-context learning** that enables compositional reasoning while outputting executable low-level actions.

3.1. Problem Formulation

Consider a robot manipulation setting where we have access to a demonstration library $\mathcal{D}^{\text{seen}}$ collected from a set of seen tasks $\mathcal{T}^{\text{seen}}$. Each demonstration $d \in \mathcal{D}^{\text{seen}}$ consists of a task instruction ℓ_d , a sequence of keyframe observations $(o_0^d, o_1^d, \dots, o_K^d)$ extracted based on gripper state changes and motion stopping criteria, and the corresponding 7-DoF actions $(a_1^d, \dots, a_K^d)$. Each action a_t specifies discretized end-effector position (3D voxel indices), rotation (Euler angles discretized into bins), and binary gripper state.

Given an unseen task $\tau^{\text{unseen}} \notin \mathcal{T}^{\text{seen}}$ with instruction ℓ^q and initial observation o_0^q , our goal is to generate an action sequence $\{a_1^q, \dots, a_T^q\}$ that successfully accomplishes the task. The key challenge lies in the distribution shift: unseen tasks may involve novel object configurations, different goal specifications, or new combinations of primitive skills not directly observed during training.

3.2. Atomic Skill Collection

To enable compositional reasoning, we construct atomic skill labels for each demonstration in the seen task library, transforming opaque numerical action sequences into interpretable skill-action pairs. As illustrated in Figure 3, our pipeline extracts keyframes from demonstrations, annotates each segment with atomic skill labels, and produces a diverse vocabulary of reusable skills.

Keyframe Extraction. Given a demonstration episode, we identify keyframes using: (1) gripper state changes (open to closed or vice versa); (2) motion stopping events, detected when joint velocities fall below a threshold; and (3) episode termination. This produces segmented demonstrations where each segment (o_k, o_{k+1}, a_k) represents a coherent atomic action.

VLM-based Annotation. For each action segment, we employ a vision-language model to annotate it with an atomic skill label following the format `Verb[obj]` or `Verb[obj1, obj2]`, where:

$$\text{Verb} \in \mathcal{V} = \{\text{Reach}, \text{Move}, \text{Grasp}, \dots\} \quad (1)$$

The VLM receives the start and end keyframe images (o_k, o_{k+1}) , gripper state transition (g_k, g_{k+1}) , object names from segmentation masks, and task instruction.

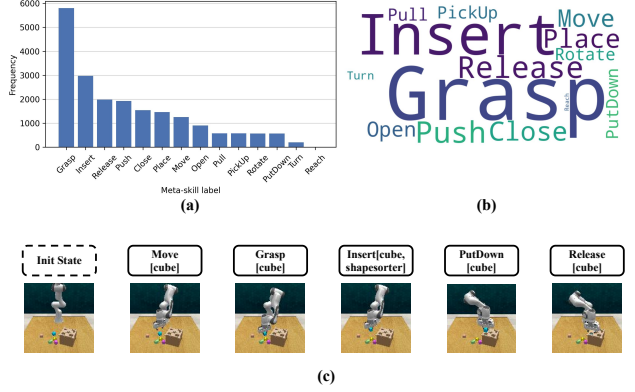


Figure 3. Atomic Skill Collection. (a) Frequency distribution of extracted atomic skills across seen demonstrations. (b) Word cloud of the atomic skill vocabulary. (c) Visualization of keyframe detection and atomic skill labeling for a sample demonstration, showing the progression from initial state through intermediate keyframes with corresponding skill annotations.

Gripper-Constrained Labeling. We leverage gripper transitions as hard constraints: when the gripper changes from open to closed ($g_k = 1, g_{k+1} = 0$), the label is forced to be `Grasp[.]`; when it changes from closed to open, the label is `Release[.]`. This ensures physical consistency and reduces annotation errors.

Post-processing. We apply rule-based post-processing: (1) for relational verbs like `Close`, `Insert`, `Place`, we enforce argument order as (movable_object, target_object); (2) for `Grasp/Release`, we verify the object belongs to the movable category; (3) when the gripper remains open during a relational action, we downgrade to `Move`. This produces clean label-action pairs $\{(s_k, a_k)\}_{k=1}^K$.

3.3. Dual-Library Demonstration Retrieval

We address the challenge of selecting demonstrations that are both relevant and skill-complete through a dual-library system: a dynamic library for task-adaptive retrieval and a static library for skill coverage completion.

3.3.1. DYNAMIC DEMONSTRATIONS LIBRARY

The dynamic library retrieves task-relevant demonstrations by fusing visual similarity with plan-based skill matching. For visual similarity, we extract a global feature from the query observation o^q using a pretrained DINOv3 (Siméoni et al., 2025) encoder: $\mathbf{f}^q = \phi_{\text{vis}}(o^q) / \|\phi_{\text{vis}}(o^q)\|_2$, and compute cosine similarity $s_i^{\text{vis}} = \mathbf{f}^q \cdot \mathbf{f}_i$ with each demonstration d_i .

For plan-based similarity, a planning agent π_{plan} first generates a predicted atomic skill sequence $\hat{\mathcal{S}} = (\hat{s}_1, \dots, \hat{s}_M)$ based on the instruction and scene state, e.g., discretized object coordinates and gripper state, and more details in Appendix A. We then measure the alignment between the

predicted plan and each demonstration’s skill labels $\mathcal{S}_i$ using Jaccard similarity over both verb sets and bigram chains:

$$s_i^{\text{plan}} = \lambda \cdot J(\mathcal{V}(\hat{\mathcal{S}}), \mathcal{V}(\mathcal{S}_i)) + (1 - \lambda) \cdot J(\mathcal{B}(\hat{\mathcal{S}}), \mathcal{B}(\mathcal{S}_i)) \quad (2)$$

where $\mathcal{V}(\cdot)$ extracts verb sets and $\mathcal{B}(\cdot)$ extracts verb bigrams. The final ranking score fuses both similarities as $s_i = \alpha \cdot \tilde{s}_i^{\text{vis}} + (1 - \alpha) \cdot s_i^{\text{plan}}$, where $\tilde{s}_i^{\text{vis}}$ is min-max normalized. We select the top- k_{sim} demonstrations to form $\mathcal{D}_{\text{dyn}}$.

3.3.2. COVERAGE-AWARE DEMONSTRATIONS LIBRARY

While the dynamic library provides task-relevant demonstrations, it may still lack critical skill patterns required for task completion. To ensure comprehensive skill coverage, we maintain an offline-constructed Coverage-aware Demonstrations Library $\mathcal{L}_{\text{cov}}$ that supplements missing skill patterns through object-agnostic coverage tokens and IDF-weighted selection.

Each demonstration is represented by tokens that abstract away object identities, capturing only the verb-level skill structure:

$$\begin{aligned} \mathcal{T}(d) = & \{V : v \mid v \in \mathcal{V}(\mathcal{S}_d)\} \\ & \cup \{B : v_1 \rightarrow v_2 \mid (v_1, v_2) \in \mathcal{B}(\mathcal{S}_d)\} \end{aligned} \quad (3)$$

To prioritize rare but potentially critical skills, we assign IDF-based weights to each token: $w_t = (\log \frac{N+1}{\text{df}(t)+1} + 1)^\beta$, where N is the total number of demonstrations, $\text{df}(t)$ is document frequency, and β controls the weighting intensity. The selection score balances coverage gain against demonstration length:

$$\text{Score}(d) = \frac{\sum_{t \in \mathcal{T}(d) \setminus \mathcal{C}} w_t}{1 + \gamma \cdot |\mathcal{S}_d|} \quad (4)$$

where $\mathcal{C}$ denotes currently covered tokens and γ is a length penalty.

During inference, we compute the coverage gap $\mathcal{G} = \mathcal{T}(\hat{\mathcal{S}}) \setminus \bigcup_{d \in \mathcal{D}_{\text{dyn}}} \mathcal{T}(d)$ between the predicted plan’s required tokens and those already covered by the dynamic library. We then greedily select up to k_{cov} demonstrations from $\mathcal{L}_{\text{cov}}$ to fill missing tokens, yielding the final demonstration set $\mathcal{D} = \mathcal{D}_{\text{dyn}} \cup \mathcal{D}_{\text{cov}}$.

3.4. Skill-Augmented In-Context Learning

Retrieved demonstrations are formatted with skill–action alignment:

$$[\text{instr}_i, \text{obs}_i, \text{plan}_i] \rightarrow [(s_1, a_1), \dots, (s_K, a_K)] \quad (5)$$

The skill labels serve as interpretable intermediate representations explaining what each action accomplishes and how it connects to the task goal. For the query task, we present

instruction, observation, and the suggested plan from the planning agent.

Crucially, while labels guide understanding of action semantics and execution order, the LLM outputs only executable actions without labels. This ensures: (1) direct compatibility with the robot controller; (2) implicit skill-level reasoning even when exact decomposition is uncertain; (3) generalization to novel skill combinations.

The system prompt emphasizes that demonstrations include skill annotations for reasoning guidance, but the model should output only action sequences. Through this formulation, our approach elevates cross-task transfer from trajectory similarity to composable skill structures, explicitly activating reasoning about skill composition rather than superficial pattern matching.

4. Experiments

4.1. Implementation Details

We evaluate our approach for zero-shot cross-task robotic manipulation generalization across two settings: the AGNOSTOS benchmark, which is the first simulation testbed specifically designed for cross-task zero-shot manipulation evaluation, and a real-world experimental setup using a UFACTORY xArm6 robotic arm. The hyperparameters λ , α , β , and γ in Dual-Library Demonstration Retrieval are 0.7, 0.5, 0.5, and 0.03, respectively. We use 20 in-context demonstrations in total, with $k_{\text{sim}} = 17$ dynamically retrieved demonstrations and $k_{\text{cov}} = 3$ coverage-aware demonstrations. We primarily employ Qwen2.5-VL as the planning agent in offline Atomic Skill Collection, and we use the Qwen-2.5-7B-Instruct model as the planning agent and LLM in the reasoning phase. These models are deployed on two A6000 GPUs.

4.2. Main Results

To systematically assess the performance of our method, we conduct experiments on the AGNOSTOS benchmark (Zhou et al., 2025), which comprises 23 unseen tasks organized into two difficulty tiers: Level-1 and Level-2.

Baseline Methods. We compare against diverse Vision-Language-Action (VLA) approaches spanning four distinct categories:

(1) Foundation VLA Models. This category encompasses models that leverage either large-scale cross-embodiment robotic datasets or are architecturally grounded in pre-trained LLMs/VLMs. These approaches aim to achieve broad manipulation capabilities through scale and language-vision grounding. Representative methods include OpenVLA (Kim et al., 2024), which pioneers open-source VLA

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Table 1. Cross-task zero-shot manipulation performance on 23 unseen tasks from the AGNOSTOS benchmark. We report success rates (%) for each task, grouped by difficulty level (Level-1 and Level-2). Column headers show task abbreviations (full names in Appendix). N/A indicates tasks overlapping with training data. Best results are in **bold**.

Category	Method	Level-1 Tasks												
		Toilet	Knife	Fridge	Micro.	Laptop	Phone	Seat	LampOff	LampOn	Book	Umb.	Grill	Bin
In-Domain	PerAct	0.0	5.3	37.3	64.0	2.7	0.0	72.0	0.0	1.3	0.0	1.3	8.0	54.7
	RVT	0.0	2.7	50.7	26.7	50.7	2.7	40.0	0.0	1.3	0.0	1.3	0.0	6.7
	Sigma-Agent	0.0	9.3	56.0	9.3	30.7	1.3	65.3	1.3	0.0	0.0	0.0	1.3	4.0
	RVT2	0.0	1.3	0.0	17.3	42.7	1.3	62.7	2.7	1.3	0.0	1.3	5.3	34.7
	InstantPolicy	0.0	1.3	13.3	4.0	4.0	18.7	24.0	0.0	0.0	0.0	0.0	0.0	0.0
Human-Video	D4R	0.0	8.0	32.0	30.7	24.0	0.0	65.3	20.0	4.0	0.0	0.0	0.0	0.0
	R3M	0.0	0.0	37.3	22.7	25.3	1.3	62.7	6.7	4.0	0.0	0.0	0.0	0.0
	D4R-Align	0.0	2.7	45.3	74.7	24.0	0.0	41.3	0.0	0.0	1.3	0.0	0.0	0.0
	R3M-Align	0.0	4.0	49.3	25.3	21.3	0.0	49.3	0.0	5.3	0.0	0.0	1.3	1.3
Foundation VLA	OpenVLA	0.0	5.3	38.7	40.0	57.3	0.0	53.3	12.0	1.3	1.3	0.0	10.7	0.0
	RDT	0.0	0.0	46.7	13.3	14.7	0.0	50.7	0.0	0.0	1.3	0.0	8.0	0.0
	π_0	0.0	5.3	85.3	24.0	40.0	1.3	64.0	18.7	8.0	1.3	0.0	33.3	1.3
	LLARVA	0.0	0.0	12.0	0.0	6.7	0.0	40.0	0.0	0.0	0.0	0.0	0.0	0.0
	3D-LOTUS	0.0	6.7	N/A	N/A	N/A	0.0	6.7	0.0	0.0	0.0	0.0	13.3	5.3
	3D-LOTUS++	0.0	5.3	N/A	N/A	N/A	9.3	68.0	10.7	0.0	0.0	0.0	29.3	13.3
	SAM2Act	0.0	0.0	36.0	40.0	6.7	6.7	62.7	6.7	0.0	1.3	1.3	9.3	0.0
	VoxPoser	0.0	0.0	0.0	0.0	5.3	8.0	28.0	88.7	25.3	0.0	0.0	0.0	82.7
ICL-based	X-ICM	1.3	26.7	22.7	45.3	33.3	57.3	48.0	58.7	50.7	1.3	0.0	8.0	18.7
	Ours	1.3	21.3	34.7	62.7	34.7	42.7	72.0	67.0	52.3	1.3	0.0	12.0	20.0

Category	Method	Level-2 Tasks										Average		
		USB	Lid	Plate	Ball	Scoop	Rope	Oven	Buzz	Plants	Charger	L-1	L-2	All
In-Domain	PerAct	58.7	2.7	0.0	0.0	0.0	0.0	1.3	4.0	6.7	2.7	19.0 $\pm$ 1.4	7.6 $\pm$ 1.1	14.0 $\pm$ 0.9
	RVT	89.3	2.7	0.0	0.0	0.0	0.0	4.0	8.0	5.3	4.0	14.0 $\pm$ 1.4	11.3 $\pm$ 1.6	12.8 $\pm$ 0.2
	Sigma-Agent	88.0	0.0	0.0	0.0	0.0	0.0	4.0	8.0	5.3	1.3	13.7 $\pm$ 1.6	10.7 $\pm$ 1.7	12.4 $\pm$ 0.4
	RVT2	22.7	40.0	0.0	0.0	0.0	0.0	0.0	1.3	1.3	1.3	13.1 $\pm$ 0.4	6.7 $\pm$ 1.3	10.3 $\pm$ 0.6
	InstantPolicy	26.7	1.3	0.0	0.0	0.0	0.0	0.0	1.3	0.0	0.0	4.3 $\pm$ 4.2	2.9 $\pm$ 1.4	3.7 $\pm$ 3.0
Human-Video	D4R	98.7	0.0	0.0	0.0	0.0	0.0	1.3	1.3	1.3	4.0	14.1 $\pm$ 0.3	10.7 $\pm$ 0.2	12.6 $\pm$ 0.2
	R3M	48.0	0.0	0.0	0.0	0.0	0.0	8.0	2.7	2.7	1.3	12.3 $\pm$ 1.4	6.3 $\pm$ 0.9	9.7 $\pm$ 0.6
	D4R-Align	89.3	1.3	0.0	0.0	0.0	0.0	8.0	6.7	0.0	1.3	14.5 $\pm$ 1.0	10.7 $\pm$ 0.2	12.8 $\pm$ 0.6
	R3M-Align	90.7	0.0	1.3	0.0	0.0	0.0	2.7	13.3	4.0	0.0	12.9 $\pm$ 0.7	11.2 $\pm$ 0.7	12.2 $\pm$ 0.3
Foundation VLA	OpenVLA	77.3	0.0	0.0	0.0	0.0	0.0	6.7	5.3	2.7	0.0	16.9 $\pm$ 1.3	9.2 $\pm$ 0.7	13.6 $\pm$ 0.8
	RDT	100.0	29.3	4.0	0.0	0.0	0.0	8.0	2.7	0.0	0.0	10.4 $\pm$ 0.5	14.4 $\pm$ 0.9	12.1 $\pm$ 0.4
	π_0	97.3	0.0	1.3	0.0	0.0	0.0	14.7	5.3	1.3	0.0	21.7 $\pm$ 0.4	12.0 $\pm$ 0.9	17.5 $\pm$ 0.4
	LLARVA	24.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.5 $\pm$ 0.1	2.4 $\pm$ 0.0	3.6 $\pm$ 0.1
	3D-LOTUS	85.3	0.0	1.3	0.0	0.0	0.0	0.0	4.0	0.0	0.0	3.2 $\pm$ 0.5	9.1 $\pm$ 0.7	6.2 $\pm$ 0.5
	3D-LOTUS++	90.7	30.7	0.0	0.0	5.3	1.3	8.0	8.7	6.7	0.0	13.6 $\pm$ 1.0	15.1 $\pm$ 1.1	14.4 $\pm$ 1.0
	SAM2Act	92.0	49.3	0.0	0.0	0.0	0.0	1.3	6.7	4.0	5.3	13.1 $\pm$ 0.4	15.9 $\pm$ 1.3	14.0 $\pm$ 0.7
	VoxPoser	32.0	76.0	0.0	8.0	0.0	0.0	0.0	1.3	0.0	4.0	18.1 $\pm$ 0.4	12.1 $\pm$ 0.4	15.6 $\pm$ 0.2
ICL-based	X-ICM	98.7	20.0	6.7	9.3	0.0	6.7	16.0	2.7	5.3	4.0	28.6 $\pm$ 1.9	16.9 $\pm$ 1.3	23.5 $\pm$ 1.6
	Ours	96.0	20.3	6.7	10.0	0.0	16.0	24.0	8.0	0.0	4.0	32.5$\pm$1.5	18.5$\pm$1.0	26.4$\pm$1.3

development; RDT (Liu et al., 2024b), employing robot-specific transformers; π_0 (Black et al., 2024), combining VLM-based flow matching with multi-platform training; LLARVA (Niu et al., 2024), utilizing instruction tuning on cross-embodiment data; SAM2Act (Fang et al., 2025) and 3D-LOTUS++ (Garcia et al., 2024), leveraging advanced visual representations; and VoxPoser (Huang et al., 2023),

synthesizing composable value maps via MLLMs.

(2) Human-Video Pretrained VLA Models. These methods exploit large-scale human activity video datasets (Grauman et al., 2022; Goyal et al., 2017) to learn rich object-interaction priors, subsequently adapting these representations for robotic control through downstream finetuning.

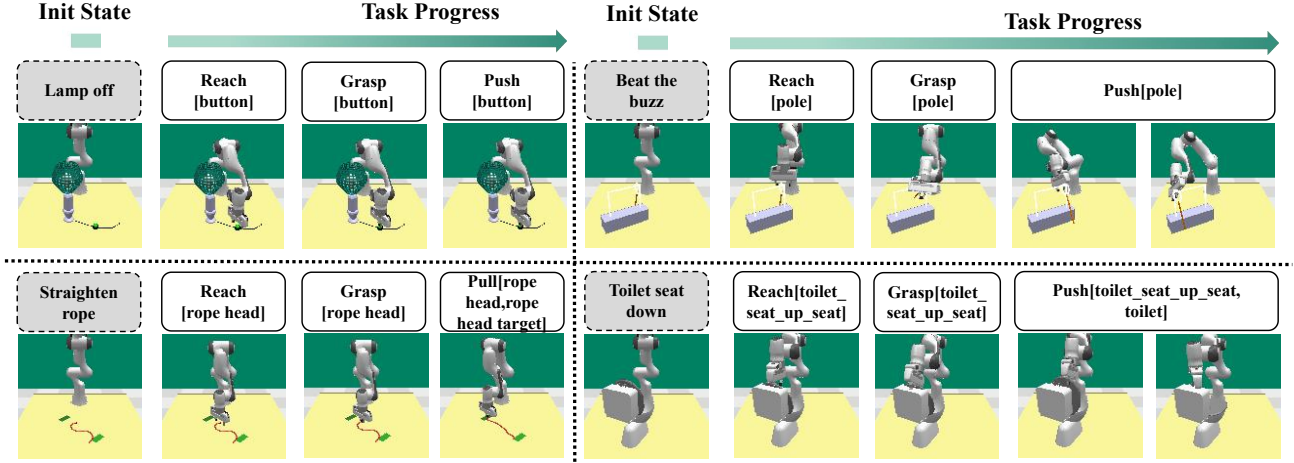


Figure 4. Visualization of our compositional skill reasoning process on four representative unseen tasks from the AGNOSTOS benchmark.

We evaluate R3M (Nair et al., 2022) and D4R (Zhou et al., 2024), along with their RL Bench-adapted variants R3M-Align (Zhou et al., 2024) and D4R-Align (Zhou et al., 2024).

(3) In-Domain Trained VLA Models. This category includes methods trained from scratch on RL Bench’s 18 standard training tasks using task-specialized architectures. These approaches serve as strong upper bounds since they operate without cross-domain distribution shift. We evaluate PerAct (Shridhar et al., 2023), RVT (Goyal et al., 2023), RVT2 (Goyal et al., 2024), Sigma-Agent (Ma et al., 2024), and Instant Policy (Vosylius & Johns, 2025)—the latter being a within-task in-context learning method based on graph diffusion over structured demonstrations.

(4) In-Context Learning Approaches for Cross-Task Manipulation. We include X-ICM (Zhou et al., 2025) as a representative method that constructs in-context prompts from seen task demonstrations to drive LLM-based action prediction for unseen tasks in a zero-shot manner.

As shown in Table 1, we conduct comprehensive evaluation across all 23 unseen tasks from the AGNOSTOS benchmark to assess both accuracy and generalization capability. For fair comparison, we follow the standard AGNOSTOS evaluation protocol, running 25 rollouts for each of 3 different seeds per task, resulting in 75 episodes per task. Our method achieves the highest overall success rates across both difficulty levels. Notably, we attain success rates exceeding 60% on four distinct tasks, including Microwave, Seat, LampOff, and USB, whereas other individual baselines achieve this threshold on at most three tasks. This demonstrates that our approach of constructing atomic skill-action alignment pairs, which preserves executable low-level actions while providing compositional and reasoned intermediate representations, effectively stimulates the model’s reasoning capabilities over skill composition and execution ordering, thereby enhancing zero-shot cross-task generaliza-

tion. Figure 4 visualizes our method’s reasoning process on four unseen tasks: `lamp_off`, `straighten_rope`, `beat_the_buzz`, and `toilet_seat_down`, providing intuitive illustration of our compositional skill reasoning for cross-task manipulation generalization.

4.3. Ablation Analysis

In Table 2, we progressively incorporate each core component to demonstrate the effectiveness of our framework. *Baseline (BS.)*: The first row presents the baseline that randomly selects in-context demonstrations from seen tasks without any retrieval strategy, achieving 21.6% overall success rate.

Dynamic Demonstrations Library (DDL.): Adding task-adaptive retrieval based on visual-semantic and skill-sequence matching improves performance to 23.3%, validating that relevant demonstration selection enhances transferable action patterns.

Coverage-aware Demonstrations Library (CDL.): Further incorporating the static library that supplements missing skill patterns raises performance to 24.9%, demonstrating the importance of skill coverage completeness.

Skill-Augmented In-Context Learning (SA-ICL.): Finally, explicit skill-action alignment annotations yield our best performance of 26.4%, confirming that structured skill representations effectively elicit compositional reasoning beyond trajectory imitation.

Furthermore, we analyze the impact of visual encoders and LLM backbones. Table 3 (upper portion) compares performance when replacing DINOv3 (Siméoni et al., 2025) with alternative general-purpose visual models such as CLIP (Radford et al., 2021) and DINOv2 (Oquab et al., 2023), demonstrating that static scene semantic extraction capability influences our method’s zero-shot cross-task

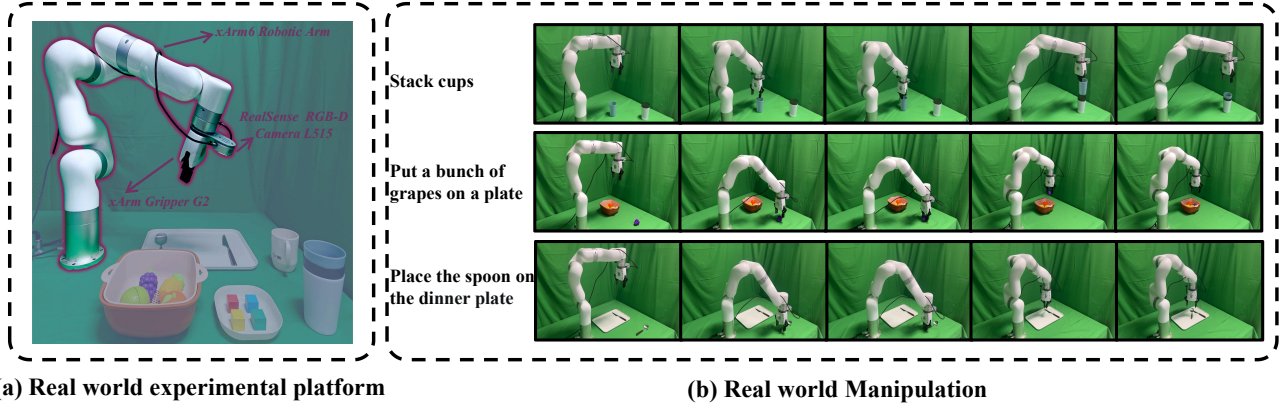


Figure 5. Visualization of our real world experiments. (a) Our real world experimental platform, consisting of a 6-DoF xArm6 arm equipped with a gripper and a RGB-D camera. (b) The results on real-world manipulation tasks.

Table 2. Ablation study on core components. BS is Baseline, DDL denotes the Dynamic Demonstrations Library; CDL denotes the Coverage-aware Demonstrations Library; SA-ICL denotes Skill-Augmented In-Context Learning.

BS.	DDL.	CDL.	SA-ICL	L-1	L-2	All
✓				26.9	14.8	21.6
✓	✓			29.5	17.2	23.3
✓	✓	✓		30.2	18.1	24.9
✓	✓	✓	✓	32.5	18.5	26.4

Table 3. Impact of visual encoders (upper) and LLM backbones (lower) on cross-task generalization performance.

Component	L-1	L-2	Overall
<i>Visual Encoder</i>			
CLIP	30.5	17.4	24.8
DINOv2	31.7	17.9	25.7
DINOv3	32.5	18.5	26.4
<i>LLM Backbone</i>			
Llama3.0-8B	28.4	14.6	22.4
Minstral-8B	24.3	14.8	20.2
InternLM3-8B	30.1	16.3	24.1
Qwen2.5-7B	32.5	18.5	26.4

generalization, underscoring the importance of scene understanding. The lower portion examines different LLM backbones: Llama3.0-8B-Instruct (Grattafiori et al., 2024), Minstral-8B-Instruct (Jiang et al., 2023), and InternLM3-8B-Instruct (Cai et al., 2024). Results reveal that backbone capability significantly impacts generalization, emphasizing that cross-task generalization improves with enhanced LLM reasoning ability. Additional ablation details are provided in Appendix B.

4.4. Real-World Experiments

To validate real-world efficacy, we conduct real-world manipulation experiments. As shown in Figure 5(a), we use a UFACTORY xArm6 arm equipped with an UFACTORY

Table 4. Real-world zero-shot cross-task manipulation results. Success rates (%) over 25 trials per task.

Task	Success Rate (%)
Stack cups	30
Put a bunch of grapes on a plate	70
Place the spoon on the dinner plate	20
Stack blocks	20
Throw the garbage into the trash can	40
Average	36

xArm Gripper G2, with RGB images and depth maps captured via a RealSense L515 camera. We provide additional details in Appendix C.

We evaluate five physical manipulation tasks: stacking cups, putting a bunch of grapes on a plate, placing the spoon on the dinner plate, stacking blocks, and throwing the garbage into the trash can. Results shown in Table 4 demonstrate our method’s zero-shot cross-task performance in real-world settings. Figure 5(b) illustrates a subset of successful cases, with additional visualization results that include both successful and failed cases can be found in Appendix C and videos are provided in the supplementary materials.

5. Conclusion

We present *Decompose and Recompose*, a skill reasoning framework for zero-shot cross-task robotic manipulation. Our core contribution is introducing atomic skill-action pairs as intermediate representations, bridging high-level task semantics and low-level controls. Through a dual-library retrieval strategy combining task-adaptive dynamic retrieval with coverage-aware static complementation, we construct skill-comprehensive demonstration sets that explicitly elicit the LLM’s compositional reasoning capabilities. Experiments on the AGNOSTOS benchmark and real-world environments validate our method’s effectiveness for cross-task generalization.

Acknowledgment

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Impact Statement

This paper presents work that aims to advance the field of Cross-Task Robotic Manipulation Generalization. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Methodology and Implementation Details

A.1. Low-level control interface

We use a standard RL Bench control interface that executes an end-effector target pose via motion planning and applies a discrete gripper open/close command. Concretely, each continuous control step is represented as

$$\mathbf{u} = [\mathbf{p}, \mathbf{q}, g] \in \mathbb{R}^3 \times \mathbb{H} \times \{0, 1\}, \quad (6)$$

where $\mathbf{p} \in \mathbb{R}^3$ is the end-effector position in the robot base frame, $\mathbf{q} \in \mathbb{H}$ is a unit quaternion specifying orientation, and g is a binary gripper command (e.g., $g=1$ for open, $g=0$ for close).

A.2. Discrete action format for the LLM

For LLM interaction, we discretize translation and rotation into integer bins and represent each action as a 7-tuple:

$$\mathbf{a} = [i_x, i_y, i_z, i_r, i_p, i_\psi, g], \quad (7)$$

where (i_x, i_y, i_z) are voxel indices for translation, (i_r, i_p, i_ψ) are discrete bins for Euler angles (roll, pitch, yaw), and g is the gripper command.

A.3. Encoding: continuous control $\rightarrow$ discrete LLM tokens

Translation discretization. Let the workspace bounds be an axis-aligned box

$$\mathbf{b}_{\min} \in \mathbb{R}^3, \quad \mathbf{b}_{\max} \in \mathbb{R}^3, \quad (8)$$

and let V be the number of uniform bins per axis (we use $V=100$). Define per-axis resolution

$$\mathbf{r} = (\mathbf{b}_{\max} - \mathbf{b}_{\min})/V. \quad (9)$$

Given a continuous position $\mathbf{p}$, the voxel index is computed as

$$\mathbf{i} = \lfloor (\mathbf{p} - \mathbf{b}_{\min}) \oslash \mathbf{r} \rfloor, \quad (10)$$

where $\oslash$ denotes elementwise division and the result is clipped to $[0, V-1]$ per axis.

Rotation discretization. We convert the quaternion $\mathbf{q}$ into Euler angles (degrees) $\boldsymbol{\theta} = [\theta_r, \theta_p, \theta_y]$ in a fixed convention. We then quantize angles with resolution Δ (we use $\Delta=5^\circ$):

$$\mathbf{k} = \lfloor (\boldsymbol{\theta} + 180^\circ)/\Delta \rfloor, \quad (11)$$

and clip each element to $[0, \lfloor 360^\circ/\Delta \rfloor - 1]$.

Gripper. The gripper command is kept discrete as $g \in \{0, 1\}$.

A.4. Decoding: discrete LLM tokens $\rightarrow$ continuous control

Given an LLM-predicted discrete action $\mathbf{a} = [\mathbf{i}, \mathbf{k}, g]$, we recover a continuous control target as follows.

Translation reconstruction. We map voxel indices to the center of the corresponding voxel cell:

$$\mathbf{p} = \mathbf{b}_{\min} + \mathbf{r} \odot \mathbf{i} + \mathbf{r}/2, \quad (12)$$

where $\odot$ denotes elementwise multiplication.

Rotation reconstruction. We recover Euler angles from bins by

$$\boldsymbol{\theta} = \Delta \cdot \mathbf{k} - 180^\circ, \quad (13)$$

then convert $\boldsymbol{\theta}$ back to a quaternion $\mathbf{q}$ using the same Euler convention as in the encoding step.

Algorithm 1 Decompose and Recompose: Skill-Based Cross-Task Manipulation**Require:** Query observation o^q , instruction ℓ^q , demonstration library $\mathcal{D}_{\text{seen}}$, Coverage-aware Demonstrations Library $\mathcal{L}_{\text{cov}}$ **Ensure:** Predicted action sequence $\{a_1^q, \dots, a_T^q\}$

```

1: // Phase 1: Plan Generation
2: Extract scene state from  $o^q$ ; generate skill sequence  $\hat{S} \leftarrow \pi_{\text{plan}}(\ell^q, \text{scene state})$ 
3: // Phase 2: Dynamic Library Retrieval
4: Encode query image:  $f^q \leftarrow \phi_{\text{vis}}(o^q) / \|\phi_{\text{vis}}(o^q)\|_2$ 
5: for each demonstration  $d_i \in \mathcal{D}_{\text{seen}}$  do
6:   Compute visual similarity  $s_i^{\text{vis}}$  and plan similarity  $s_i^{\text{plan}}$ 
7:   Fuse scores:  $s_i \leftarrow \alpha \cdot \hat{s}_i^{\text{vis}} + (1 - \alpha) \cdot s_i^{\text{plan}}$ 
8: end for
9: Select top- $k_{\text{sim}}$  demonstrations:  $\mathcal{D}_{\text{dyn}}$ 
10: // Phase 3: Static Library Coverage Completion
11: Compute covered tokens  $\mathcal{C}$  and missing patterns  $\mathcal{G} \leftarrow \mathcal{T}(\hat{S}) \setminus \mathcal{C}$ 
12: while  $|\mathcal{D}_{\text{cov}}| < k_{\text{cov}}$  and  $\mathcal{G} \neq \emptyset$  do
13:   Select  $d^* \leftarrow \arg \max_d \text{Score}(d)$  from  $\mathcal{L}_{\text{cov}}$ ; update  $\mathcal{D}_{\text{cov}}, \mathcal{C}, \mathcal{G}$ 
14: end while
15: // Phase 4: Skill-Augmented In-Context Learning
16: Merge demonstrations:  $\mathcal{D} \leftarrow \mathcal{D}_{\text{dyn}} \cup \mathcal{D}_{\text{cov}}$ 
17: Return  $\{a_1^q, \dots, a_T^q\} \leftarrow \text{LLM}(\ell^q, o^q, \hat{S}, \mathcal{D})$ 

```

Execution. Finally, the low-level control command is assembled as $\mathbf{u} = [\mathbf{p}, \mathbf{q}, g]$ and executed by the RLBench action mode (end-effector pose via planning + discrete gripper). This completes the closed loop between the LLM interface and the robot controller.

A.5. Practical notes

In practice, clipping ensures indices remain valid even when the predicted pose slightly exceeds workspace limits. Using voxel centers provides a consistent inverse mapping and reduces discretization bias. Rotation discretization is similarly bounded to guarantee valid bins, while quaternion normalization is applied after reconstruction to ensure a valid rotation.

A.6. Algorithm details

Algorithm 1 summarizes our complete pipeline. The algorithm proceeds in four phases: (1) **Plan Generation** produces a predicted skill sequence $\hat{S}$ that guides subsequent retrieval; (2) **Dynamic Demonstrations Library** selects task-adaptive demonstrations by fusing visual similarity with plan-based similarity; (3) **Coverage-Aware Completion** greedily supplements missing skill patterns from the static library using IDF-weighted scoring with length regularization; (4) **Skill-Augmented ICL** constructs the final prompt where skill labels provide compositional reasoning guidance while the model outputs only executable actions.

The dual-library design achieves complementary objectives: the dynamic library provides *task-adaptive* demonstrations based on visual and semantic relevance, while the static library ensures *skill-coverage sufficiency*. This combination enables the LLM to reason over composable skill structures rather than merely imitating trajectory patterns.

A.7. More visual results

Figure 6 illustrates the reasoning process of our method on six unseen task of the AGNOSTOS benchmark.

B. More Ablation

We conduct additional ablation studies to analyze the impact of demonstration quantity on cross-task generalization performance. Results are summarized in Table 5.

Table 5. Ablation study on the number of demonstrations (upper) and coverage-aware demonstrations (lower) in in-context learning.

Component	L-1	L-2	Overall
<i>Demos per ICL</i>			
0	0.0	0.0	0.0
5	27.3	13.1	19.8
10	30.0	15.5	23.7
15	31.2	16.8	24.9
20	32.5	18.5	26.4
<i>Coverage-aware Demos per ICL</i>			
0	31.0	17.1	24.9
1	31.6	17.7	25.5
2	32.1	18.2	26.0
3	32.5	18.5	26.4
4	32.3	18.1	26.1

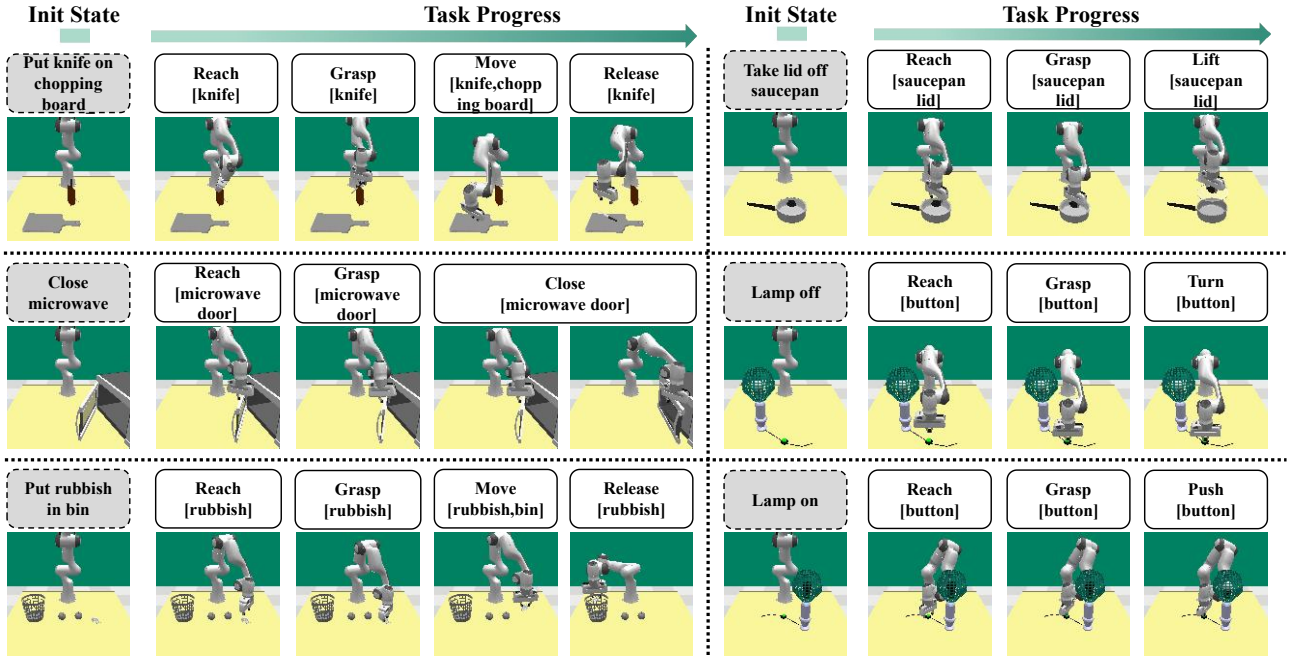


Figure 6. Visualization of our compositional skill reasoning process on some representative unseen tasks from the AGNOSTOS benchmark.

Number of in-context demonstrations. The upper portion of Table 5 examines how the total number of demonstrations in the in-context prompt affects performance. When no demonstrations are provided, the model completely fails to generate valid action sequences, achieving 0% success rate across all tasks. This confirms that in-context demonstrations are essential for cross-task manipulation, as the LLM cannot infer appropriate action patterns from task instructions alone. As the number of demonstrations increases from 5 to 20, we observe consistent improvements: overall success rate rises from 19.8% to 26.4%, with Level-1 tasks improving from 27.3% to 32.5% and Level-2 tasks from 13.1% to 18.5%. This trend indicates that richer demonstration sets provide more comprehensive skill–action patterns for the model to reason over.

Number of coverage-aware demonstrations. The lower portion of Table 5 investigates the contribution of coverage-aware demonstrations, which supplement missing skill patterns not captured by the dynamic retrieval. Starting from a baseline of 0 coverage-aware demonstrations, i.e., relying solely on dynamically retrieved examples, performance stands at 24.9%.

Adding coverage-aware demonstrations yields steady improvements: 1 demo achieves 25.5%, 2 demonstrations reach 26.0%, and 3 demonstrations attain the best overall performance of 26.4%. Notably, performance slightly decreases with 4 demonstrations, suggesting that excessive static demonstrations may introduce noise or dilute the relevance of task-adaptive examples. This indicates an optimal balance exists between ensuring skill coverage completeness and maintaining

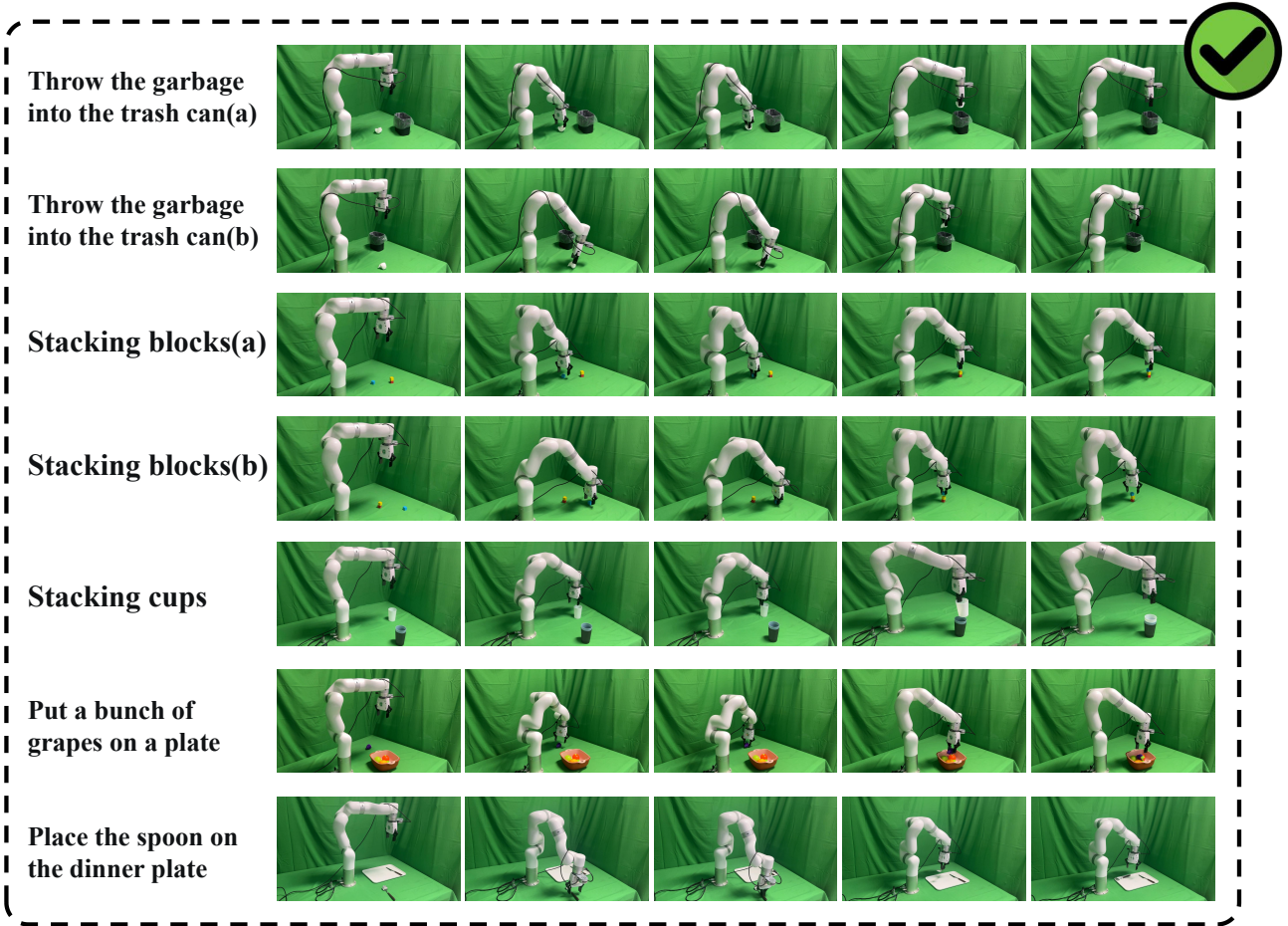


Figure 7. Successful cases of our real world experiments.

demonstration relevance.

These results validate two key design choices: (1) sufficient demonstration quantity is necessary to expose diverse skill-action patterns, and (2) the coverage-aware static library effectively complements dynamic retrieval by filling critical skill gaps, but should be used judiciously to avoid over-saturation.

C. Real-world experiments

Experimental setup: we use a UFACTORY xArm6 arm equipped with an UFACTORY xArm Gripper G2, with RGB images and depth maps captured via a RealSense L515 camera. We evaluate five physical manipulation tasks: stacking cups, putting a bunch of grapes on a plate, placing the spoon on the dinner plate, stacking blocks, and throwing the garbage into the trash can. We collect 20 demonstrations per task, and during testing, construct cross-task in-context prompts using demonstrations from the remaining four tasks, enabling zero-shot generalization. We employ Qwen2.5-7B-Instruct as the backbone with 20 seen demonstrations. Each task is executed 25 times, with average success rates reported.

Figure 7 shows successful examples from our real-world task experiments. Across all five tasks, our method demonstrates the ability to generate appropriate skill sequences and execute precise actions in diverse manipulation scenarios. For instance, in the *stack cups* and *stack blocks* tasks, the robot accurately grasps objects and places them at the correct positions. These results indicate that our framework effectively transfers skills learned from seen tasks to novel manipulation scenarios in real-world environments.

Additionally, as illustrated in Figure 8, we present several representative failure cases from our real-world experiments. In

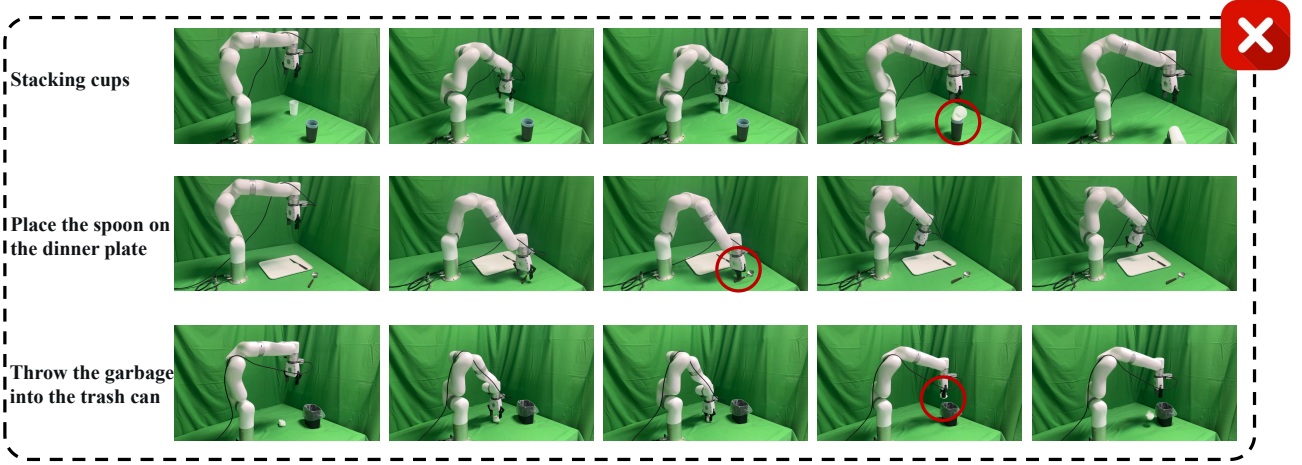


Figure 8. Failing cases of our real world experiments.

the first row, during the *stack cups* task, the robot arm successfully navigates to the stacking location; however, the cup is grasped at a tilted angle, causing the subsequent placement to fail due to misalignment. The second row depicts a failure in the *place the spoon on the dinner plate* task, where the thin handle of the spoon prevents the gripper from achieving a secure grasp. The third row shows the execution of *throw the garbage into the trash can*. In this case, although the robot successfully grasps the garbage and moves it above the trash can, the end-effector hovers over the edge rather than the center of the bin. Consequently, when the gripper releases, the garbage falls outside the container.

These failure cases highlight critical challenges for cross-task robotic manipulation. First, understanding the 6-DoF pose of real-world objects, including their geometric properties and functional affordances, is essential for reliable grasping and placement. Second, accurate reasoning about 3D spatial relationships, such as determining whether the end-effector is positioned above the center versus the periphery of a target region, remains crucial for successful task completion. In future work, we plan to address these limitations by incorporating explicit 3D spatial reasoning and object affordance understanding into our framework.